

Assignment #4

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Overview:

In this Assignment you will create Go programs that use the HTTP communication protocol and REST API for building a web service on the personal development computer.

Submission:

Submit your exercise as a SINGLE ZIP file on Canvas by the due date. Your submitted ZIP file must have the name:
Assignment_4_Your_LastName.zip

Deliverables:

Your ZIP file for the Assignment submission must include the following:

- All source code that you wrote, installed, compiled, and built on your personal development computer.
- Panopto video recording of a live run (5 to 10 minutes) of your code on your personal development computer.

Requirements Specification:

Implement the following requirements, and test-drive your Go application:

1. (15 Points) Build and run **handlers.go** and **www-phone.go** programs
2. (15 Points) Use **curl** (download from this URL: <https://curl.se/download.html>) to execute the following commands from the command window/terminal:
 - 2.1. curl localhost:1234/list
 - 2.2. curl localhost:1234/status
 - 2.3. curl localhost:1234/search/2109416471
 - 2.4. curl localhost:1234/delete/2109416471
 - 2.5. curl localhost:1234/delete
 - 2.6. curl localhost:1234/insert/Mike/Tsoukalos/2109416471
3. (15 Points) Modify and run **handlers.go** and **www-phone.go** as follows:
 - 3.1. Replace **Entry struct** with **MSDSCourse struct** that has course name, description, and prerequisite:

```
type NoEmpty struct {  
    Name  string `json:"username"`  
    Surname string `json:"surname"`  
    Year   int   `json:"creationyear,omitempty"`  
}
```

With

```
type MSDSCourse struct {  
    CID   string `json:"courseID"`  
    CNAME string `json:"course_name"`  
    CPREREQ string `json:"prerequisite"`  
}
```

- 3.2. Replace **type PhoneBook []Entry** with type **MSDSCourseCatalog []MSDSCourse**
4. (15 Points) Use **curl** (download from this URL: <https://curl.se/download.html>) to execute the following commands from the command window/terminal:
 - 4.1. curl localhost:1234/list
 - 4.2. curl localhost:1234/status
 - 4.3. curl localhost:1234/search/MSDS432

4.4. curl localhost:1234/delete/MSDS432

4.5. curl localhost:1234/delete

4.6. curl

localhost:1234/insert/MSDS432/FoundationsOfDataEngineering/MSDS420